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[21MEDE742]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER – 2024

AUTOMOBILE ENGINEERING

Time:3 Hours **Maximum Marks:50**

Instructions:
- i. PART - A (Question 1-5) answer any 4 questions.
- ii. PART - B (Question 6-8) answer any 2 questions.
- iii. PART - C (Question 9) is compulsory

PART – A 05X04=20

1.	a.	Describe the function of Variable Valve Timing (VVT) in IC engines	05
2.	a.	Differentiate between manual and automatic gearboxes.	05
3.	a.	Explain the factors affecting tire life.	05
4.	a.	What are the key safety features of airbags in vehicles?	05
5.	a.	Briefly explain the greenhouse effect and how automobile emissions contribute to it.	05

PART - B 09X02=18

6.	a.	Discuss the working principle of a torque converter with the aid of a suitable sketch	09
7.	a.	Explain the basic layout and components of electrical vehicles	09
8.	a.	Explain the different types of braking systems (pneumatic & hydraulic) and compare their efficiency.	09

PART - C 01X12=12

9.	a.	Discuss the construction and function of various components in IC engines with the help of a suitable sketch and the materials used.	12

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